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WEST BENGAL UNIVERSITY OF TECHNOLOGY

M-402

MATHEMATICS-III

Time Allotted: 3 Hours | Full Marks: 70

The questions are of equal value.
The figures in the margin indicate full marks.
Candidates are required to give their answers in their own words as far as practicable.

GROUP A
(Multiple Choice Type Questions)

1. Answer any *ten* questions. 10×2 = 20

(i) Evaluate $\oint \frac{z}{z^2-1} dz$, where $C: |z|=2$.

(ii) Find the singularities of $f(z)=\frac{1}{\sin z - \cos z}$.

(iii) The mean of Binomial distribution is 5, find the value of $E(2X-9)$.

(iv) Prove that $J_{\frac{-3}{2}}(x)=\sqrt{\frac{2}{\pi x}}(-\frac{\cos x}{x}-\sin x)$.

(v) Expand $f(z)=\frac{1}{(z-1)(z-2)}$ in Laurent's series in the region $|z|<1$.

(vi) State Cauchy's integral formula for the derivative of an analytic function.

(vii) Find the residue of $f(z)=\frac{z}{(z-1)^4(z-2)(z-3)}$ at $z=2$.

(viii) If A, B, C are independent events, prove that the events A and (B $\cup$ C) are independent.

(ix) If X is a Poisson variate such that $P(X=2) = 9P(X=4) + 90\,P(X=6)$. Find the mean and variance of X.

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(x) A random variable X has the density function

$$f(x) = x,\ 0 < x < 1,$$
$$= \frac{1}{2},\ 1 < x < 2.$$

Find the mean of X.

(xi) The probability density function of a random variable x is

$$f(x) = k(x-1)(2-x), \text{ for } 1 < x < 2.$$

Determine the value of the constant k.

(xii) Find the regular singular point(s) of the equation

$$2x^2y'' + (2x^2 - x)y' + y = 0.$$

(xiii) Define Bessel's function of 1st kind of order n, J_n.

(xiv) Prove that, $P_n(1) = 1$.

GROUP B

(Answer any *five* questions taking at least *one* from each Module.) $10\times5 = 50$

Module - I

2. (a) If $f(x)=\begin{cases} 0, -\pi \le x \le 0 \\ \sin x, 0 \le x \le \pi \end{cases}$ 5

then find the Fourier series of $f(x)$ and prove that

$$f(x) = \frac{1}{\pi} + \frac{1}{2}\sin x - \frac{2}{\pi}\sum_{n=1}^{\infty} \frac{\cos 2nx}{4n^2 - 1}.$$

(b) Find the Fourier cosine series of the function $f(x) = \pi - x$ in $0 \le x \le \pi$ and hence deduce that $\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \ldots = \frac{\pi^2}{8}$. 5

3. (a) Find a series of cosine multiples of x which will represent $f(x) = x$, $0 \le x \le 2$ and write Parseval's identity corresponding to it. Hence, show that $\frac{1}{1^4} + \frac{1}{2^4} + \frac{1}{3^4} + \ldots = \frac{\pi^4}{90}$. 3+2+1

(b) Find the inverse Fourier Transform of the function $\frac{1}{s^2 + 6s + 25}$. 4

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4. (a) Find the Fourier Transform of the function 5

$$f(x)=\begin{cases}1, \text{for} |x|\le 1\\ 0, \text{for} |x|>1\end{cases}$$

and hence show that $\int_0^\infty \frac{\sin x}{x}dx=\frac{\pi}{2}$.

(b) Find the Fourier Cosine Transform of the function $f(x)=e^{-x^2}$. 5

Module – II : Calculus of Complex Variable

5. (a) Show that $f(z)=|z|^2$ is differentiable at the point $z = 0$, but not analytic there. 3

(b) Prove that the function 7

$$f(z)=\begin{cases}\frac{x^3(1+i)-y^3(1-i)}{x^2+y^2}, & z\neq 0\\ 0, & z=0\end{cases}$$

is continuous and the Cauchy-Riemann equations are satisfied at the origin, yet $f(z)$ is not differentiable at origin.

6. (a) Prove that the function $u(x, y) = x^3 - 3xy^2 + 3x^2 - 3y^2 + 1$ is harmonic. Find the analytic function in terms of z. 5

(b) $\int_C \frac{z+1}{z^2-2z}dz$, where C is the circle $|z|=5$. 5

7. (a) Find the Laurent's series for $f(z)=\frac{1}{(z+1)(z+3)}$ valid in: 5

(i) $1<|z|<3$

(ii) $0<|z+1|<2$

(b) Evaluate $\int_0^{2\pi}\frac{\cos 2\theta}{5+4\cos\theta}d\theta$, using Cauchy's residue theorem. 5

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Module – III : Probability

8. (a) The probability that a doctor diagnoses a patient correctly is 80%. The probability that a patient dies after correct diagnosis is 40% and the probability that he dies after wrong diagnosis is 70%. The patient died. What is the probability that the disease was correctly diagnosed? 5

   (b) The radius of a circle has distribution given by the p.d.f. 5

$$f(x)=\begin{cases}1, 1<x<2\\ 0, \text{ otherwise}\end{cases}$$

   Find the mean and variance of the area of the circle.

9. (a) A point X is chosen at random on a line segment AB of length $2a$ with mid-point O. find the probability that AX, BX and AO can form the sides of a triangle. 4

   (b) 50% of a product have weight 60 Kg of more and 10% have weight less than 55 Kg. Assuming the distribution of weight to be normal (μ, σ^2), evaluate $(\mu - 3\sigma, \mu + 3\sigma)$. For a standard normal variate Z, $P(Z \leq 1.28) = 0.90$ approximately. 6

10.(a) In a bolt factory, machines A, B and C manufacture respectively 25%, 35% and 40% of the total of their output. 5%, 4% and 2% are defective bolts. A bolt is drawn at random from the product and is found to be defective. What is the probability that it is manufactured by the machines A, B and C? 6

   (b) If A and B are events with $P(A) = 3/8$, $P(B) = 5/8$ and $P(A \cup B) = 3/4$, find $P(B|A)$ and $P(A|B)$. Are A and B independent? 4

Module- IV : PDE and Series Solutions of ODE.

11.(a) State and prove Rodrigue's formula. 6

   (b) Express $P(x) = x^4 + 2x^3 + 2x^2 - x - 3$ in terms of Legendre polynomials. 4

12. Apply the method of separation of variables to find the solution of the wave equation 10

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}.$$

Given that,

$$u(0,t)=u(l,t)=0,$$

$$u(x,0)=f(x),\ \frac{\partial u}{\partial t}(x,0)=0,\quad 0<x<l.$$